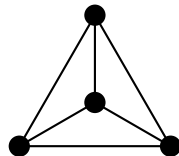


The First Distinction

A Companion to Void and Form

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Built with AI

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Opening

Why is there anything at all, rather than nothing?

Since Leibniz, this question has accompanied every inquiry into origin. In modern cosmology it appears again: the closer the description of the universe moves toward its beginning, the more clearly a limit comes into view. That limit is often read as a consequence of missing data. Yet the question of origin may touch a deeper level.

And here a remarkable structure shows itself. The very formulation of the question already presupposes a separation. *Something* and *nothing* appear as distinguishable possibilities. A nothing that cannot be distinguished from something cannot be asserted as nothing at all. The pressure of the question thus already produces the distinction to which it points. The question of origin therefore leads immediately to the more general question of how distinction becomes possible at all.

Spencer Brown observed that every description begins with a distinction drawn. Wheeler suspected that information is not posterior to the physical world but prior to it. Both thoughts direct attention to the conditions under which anything can be described, observed, or known at all.

This book follows that trail.

Its starting point is the conjecture that every piece of information rests on a distinction successfully carried out. Before anything can be measured, classified, or explained, a difference must be brought forth and held stable. The question of origin therefore leads not only toward physics, but also toward the minimal performances that must already have been achieved if information is to arise at all.

At the center of the inquiry, then, stands a process. The guiding question is this: what minimal structure must be in place so that a distinction does not collapse, remains complete, and allows stable information to emerge from it?

This work follows that question to its first formally graspable threshold. It studies the minimal structure that preserves the trace of a successful act of distinction and formalizes that threshold in intensional Martin-Löf type theory, machine-checked in Agda.

The formal core thus appears not as the starting point of the inquiry, but as its first precise footprint. From there the connection is traced back to the question of origin: to information, observation, space, time, and the conditions of physical description.

This book belongs to a long line of questions, running from Leibniz through Spencer Brown to Wheeler. Its ambition is to carry a small part of that common pressure into formal shape and thereby make visible what must already have happened if anything at all is to be describable.

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Chapter 1

The Question of Origin

“Why is there something rather than nothing? For nothing is simpler and easier than something.”

Gottfried Wilhelm Leibniz

In 1714 Leibniz gave the question of origin its sharpest form: why is there something rather than nothing? Its sharpness lies in a simple observation. Nothing, Leibniz remarks, is simpler and easier than something. If the world could just as well have been empty, then precisely its not being empty demands an account.

No measurement answers this question. One can trace the history of the cosmos back to the first physically describable state and still leave open why there is a cosmos whose history can be traced at all. The question reaches beyond everything data alone can decide.

But it already carries its own condition within itself. To ask between *something* and *nothing*, both must already be in force as different. A nothing that could not be distinguished from something could not appear as nothing. The question of origin therefore already presupposes a distinction it does not itself explain.

That is why it leads immediately further. Not only: why is there something rather than nothing? But also: what must already hold for something to be distinguishable from nothing at all?

Chapter 2

What Every Determination Already Presupposes

If the question of origin really hangs on a distinction, then the next question follows at once: does this hold only for that one question, or wherever something appears as this and not as that?

The suspicion can be sharpened at the simplest case.

A mark is this mark and not the blank place on the page. This sheet is this sheet and not that one. One side is this side and not the other. Wherever something is determinate, something else is excluded.

That is why one cannot retreat behind the holding-apart itself and hope to find a deeper, purer ground there. Without this elementary difference there would be only that nothing which cannot count as determinate. A sign that is not distinguished from what it refers to is no sign. A mark that does not stand out from the sheet on which it stands is no mark.

This does not yet settle everything. But the suspicion suggests itself: what became visible in the question of origin may reach further than that one question. Wherever something appears as this and not as that, something may already have been set off from something else. What became visible in Leibniz through the contrast between something and nothing may therefore concern not only the question of origin.

If that is so, it should show itself most clearly where this pressure has been driven furthest: in physics. Does physics encounter here only a limit of our knowledge, or a condition for anything to appear as different at all?

Chapter 3

The Edge of the Picture

If that boundary is real, physics is where it should be hardest to ignore — the place where the attempt has been driven furthest to grasp the beginning of the universe as a state of the world.

Modern cosmology reconstructs the history of the universe backward. Galaxies move closer together, the universe becomes hotter and denser, atoms dissolve into plasma, nuclei break apart. This reconstruction is quantitative, tested, and in important regions extraordinarily precise. Yet it does not reach its first state. The closer physics comes to the beginning, the thinner the data become. Light does not reach us from before a certain point. Energies climb beyond everything an experiment can reproduce. Space and time cease to behave like magnitudes that could simply be carried further within the same physical theory.

One can take this to be merely a lack of data. Perhaps, so the natural expectation goes, deeper measurements or a deeper theory will suffice. But this expectation already carries an assumption of its own: that better data will in the end also yield better answers about origin.

The history of physics speaks cautiously against that hope. Newton brought celestial mechanics and falling bodies under a common law, but presupposed space, time, and mass. Maxwell bound electricity, magnetism, and light into a few equations, but presupposed charge and the continuum in which it appears. Einstein made gravitation into the geometry of spacetime, but took matter and coupling constants as given. Quantum field theory yields the most precise predictions we have, but in turn requires a symmetry group, a particle inventory, and constant parameters that it does not generate from itself.

This is not an accident of individual theories. It is a recurring form. Each

theory reaches a point at which it explains more than its predecessors and yet must presuppose something it does not itself produce.

The same result can be reached without this history. Data are always data of something: states, events, relations, measured values. For them to be data at all, something must be distinguished from something else. The beginning of the universe, however, is not simply one more state within an already differentiated world. It concerns the question of how such a world can come into force at all.

The first moment is therefore not only an unsolved problem of physics. It marks the place where physics touches its own presupposition. Physics works with distinguishable states. Remove that difference, and no measurement, no dynamics, no information, and no theory remain.

The question thereby turns around. Not: which state was the first? But: what must hold for states to be in force at all? As soon as this becomes visible, the issue is no longer only the universe. It is the condition under which anything can appear as different at all. The suspicion from the question of origin therefore returns here on physical ground. But if that is so, why does information itself depend on distinction?

Chapter 4

Why Information Needs Distinction

Once the boundary no longer appears merely as a peculiarity of cosmology, the next question comes into view: why does information depend on distinction?

A measured value is this value only because it is not another. A recorded bit is this bit only because a different outcome remains excluded. The direction of the suspicion is visible here, though not yet complete.

Spencer Brown placed at the beginning of his calculus an instruction: draw a distinction. Everything else grows from that single act. Wheeler formulated his suspicion from the other side: *it from bit*. Whatever occurs physically depends on answers to yes-or-no questions, and thus on differences held in place.

Set beside one another, the two thoughts sharpen the same suspicion. A distinction drawn and a bit recorded may be the same event read twice: once as an act, once as a retained difference. Then the step that appeared in Leibniz as the pressure of the question, and in physics as the limit of states, appears here once more in another language. If that is so, then without a difference held in place there would be no information.

This book takes that as its working conjecture. Every piece of information rests on a successful act of distinction. The weight falls on *successful*. A difference that disintegrates at once carries nothing. A difference that cannot stand out from neighboring possibilities carries nothing. For information to arise, a difference must be produced, secured against collapse, and held completely enough to present itself again as the same.

The inquiry therefore shifts from boundary to performance. If information

rests on a successful act of distinction, what must that act be able to do? What is asked now is the minimal performance itself: can a difference be stated at all? Does it not fall back at once into one? And can a field of such differences be held completely, with neither less nor more?

Chapter 5

The First Demand: Distinguishability

It helps, at this point, to stop treating a distinction as a thing and start treating it as something done. What we are following is an act: the holding-apart of one from another and the maintenance of that difference. To ask what distinction requires is to ask what such an act needs in order to succeed at all.

A mark on a page counts as a sign only because it stands out from the page — because something is read *as* mark and the rest *as* background. A measurement counts as a value only because it could have come out differently: this value and not that one. In neither case is anything yet being counted. There is only the most elementary performance, without which nothing further can begin: that something can appear as this rather than that, and that this appearing holds.

This sharpens the first demand. Before anything can be counted, ordered, or measured, there must somewhere be a difference that can make itself count — the mere availability of this rather than that. Call that distinguishability.

And one must see how little this deliberately is. It is not yet counting: nothing in it says how many positions there are, or that they appear in any fixed number at all. It is not yet structure: nothing says how the sides are arranged, or which is which. It is not even yet stability: nothing says that the difference, once stated, can be found again. It is only the raw availability of a contrast, this set against that, holding so long as it is attended to. Everything the book later builds — Non-Collapse, plurality, and at last a determinate number — is laid on top of this. None of it can be had without distinguishability; distinguishability can be had without any of it. That is what it means to come first. But the matter is not yet carried. A

once-indicated difference could collapse again at once. What is needed so that it does not immediately fall back into one?

Chapter 6

The Second Demand: Non-Collapse

What distinguishability leaves open is whether a difference holds as a difference at all. Something can lie apart for a moment and in the next fall back into one. Then the difference carries nothing. No mark would hold anything. No measured value could rest on it. No state could be asserted against another.

That is why a second demand follows distinguishability. A difference does not suffice merely because it flashes up once. It must be secured against collapse. Call this demand Non-Collapse.

It is more fundamental than it first sounds. Long before there are scales, observers, or instruments, it must be possible for a difference not to vanish at once. The limit of this second demand is therefore visible as well: a difference can appear and yet fail to hold. Measurement does not create that. Recording does not create it. Both already presuppose it.

Only after that does the path lead back to physics. The same demand appears there at a prominent point: in the black-hole information problem. The issue there is not first of matter or energy, but of whether differences once achieved can be lost without remainder. This is not a proof of the present argument. But it does show the same structural demand in a physical case.

The demand can now be sharpened. Non-Collapse means: there must be a genuine separation between two positions, and that separation may not fall back into identity. That is exactly why the formal part will later have to work with inequality and equality.

With this much in hand, the next question arises. Is a single difference se-

cured against collapse enough, or must an entire field of such differences stand complete?

Chapter 7

The Third Demand: Exact Plurality

So we reach the last of the three steps, the one the whole road has been climbing toward. The question is no longer whether a difference can be stated or recovered, but what it takes for a whole field of differences to stand exactly: not one difference that flashes up and vanishes, but a standing field of them, each held as itself. This is plurality — and the question that has shadowed us from the first page can now be stated in its sharpest pre-formal form. What does it take for a system to be fixed to a determinate *many*: to the claim that here are exactly these positions, neither fewer nor more?

The demand is felt most directly if one tries to hold a small plurality in place and watch what can go wrong. Lay a few blank sheets on a table and claim that there is a definite number of them. Two failures lie in wait, and they pull in opposite directions. The first: some of the sheets counted as different might not really be different at all — if every way of pointing to one points equally well to another, then there are not distinct sheets, but one sheet pointed at twice. Call this collapse: the many secretly fall together into fewer. The second, less obvious case: there may be more on the table than the count ever mentioned — a sheet at the edge that the accounting never ruled out. Call this surplus: the field hides elements the count never closed.

These two dangers are distinct. Collapse means fewer than claimed. Surplus means more than claimed. Whoever prevents only collapse may still have too much in the field. Whoever excludes only surplus may still secretly identify two positions. An exact plurality has to answer both dangers at once.

The demand that answers both has exactly two parts. The named positions must really lie apart. Call that *separation*. And every position of the field must be captured by the account. Call that *cover*. Separation prevents collapse. Cover prevents surplus. Only both together turn named positions into an exact plurality.

At this exact point something changes, and it is worth marking rather than letting it pass. Up to here the conditions have been worked out in ordinary prose. With separation and cover, there is for the first time a specification complete enough to be written down exactly.

And now, for the first time, a determinate number steps forward. Ask where these two conditions can first be met completely, and the answer is: on the smallest field on which there is anything to hold apart at all, namely a field of two positions. With two, and not before, separation and cover can be written down completely and finitely. Two is not the origin of distinctness. But it is the first place at which its general conditions become fully explicit.

Now the question changes once more. If this is the minimal performance that stable information requires, how little must be granted to write it down exactly?

Chapter 8

How Little Can Be Granted?

The previous chapter narrowed the question to something small and exact — small enough, at last, to be written down completely. A target that precise calls for a precise instrument, and for one of an unusual kind: a language in which writing something down and checking it are one and the same act.

There is such a language. The one used here is Martin-Löf type theory in the form realized by the proof assistant Agda. In it, to assert is to construct. A statement is a type, a proof is a term inhabiting that type, and a machine decides whether the term truly has the type claimed. Nothing is accepted on authority, and nothing on fluency. A proof that does not check is not a proof one has to argue over; it is a file that does not compile. We arrive here, rather than in informal mathematics, for a single reason: not because type theory is the only conceivable carrier of the argument — other formal systems might carry the same content perfectly well, and whether they do remains open until it is shown — but because it makes the cost of every step explicit and refuses to move forward on credit.

For our purposes, that refusal is exactly the discipline required. Everything the argument uses has to be declared, and two settings, fixed at the head of the file before a single line of mathematics appears, govern everything that follows. The flag `--safe` closes every escape hatch: no postulates, no principle asserted without proof, no back door through which an unearned result could enter. The flag `--without-K` withholds a convenient but additional assumption about equality, so that nothing in the development rests on more than the bare identity type. Under them, only the things we build exist, in full view.

What we are prepared to grant can now be said in a single breath: a carrier with two positions, a witness that the two are not equal, and an account that leaves

no point of the carrier unaddressed. No numbers, no geometry, no physics, and no classical logic beyond what we construct by hand. Whatever the threshold yields will have been bought with exactly this and nothing else — and only in that way can the result honestly be credited to distinction itself rather than smuggled in through a richer setting.

What now follows is neither a technical appendix nor a change of subject. It is the same movement under a harsher discipline. The previous chapters named in ordinary language the performances a stable distinction must carry. The file now writes those performances down one by one in a language where hidden shortcuts are not allowed.

Before distinction can be written down, the few logical means on which it rests must explicitly be in place, and each of them is built rather than assumed. The order of writing is not the same as the order of the inquiry. By the time the formal file begins, the argument has already reached distinction, Non-Collapse, and exact plurality in ordinary language. A checker, however, cannot rely on what prose lets run in the background. Some capacities that have been presupposed throughout the described process now have to be written out explicitly. The file therefore begins by importing only the bookkeeping of universe levels — the type theory’s internal device for keeping its own sizes consistent. Nothing mathematical is imported with it.

```
open import Agda.Primitive using (Level; lzero; lsuc; _⊔_; Setω)
```

That first line is only housekeeping. It gives the checker a way to track sizes consistently, and none of the conceptual weight of the argument rests on it. The real formal work begins with the thinnest device needed to say when something is the same and when it is not.

Equality therefore comes first in the file, not because sameness would be prior to distinction in the order argued by this book, but because once a distinction is written down exactly, the checker needs a minimal way to say same, not the same, and substitution along sameness. It is the identity type: a family with a single constructor, `refl`, witnessing that a thing is equal to itself. From this one constructor the familiar operations — symmetry, transitivity, congruence under arbitrary functions, and substitution of equals for equals — follow by pattern matching, with no additional principle added.

```
infix 4 _≡_
```

```
data _≡_ {ℓ : Level} {A : Set ℓ} (x : A) : A → Set ℓ where
```

```

refl : x ≡ x
{-# BUILTIN EQUALITY _≡_ #-}

sym : {ℓ : Level} {A : Set ℓ} {x y : A} → x ≡ y → y ≡ x
sym refl = refl

trans : {ℓ : Level} {A : Set ℓ} {x y z : A} → x ≡ y → y ≡ z → x ≡ z
trans refl q = q

cong : {ℓ₁ ℓ₂ : Level} {A : Set ℓ₁} {B : Set ℓ₂} (f : A → B) {x y : A} → x ≡ y → f x ≡ f y
cong f refl = refl

subst : {ℓ₁ ℓ₂ : Level} {A : Set ℓ₁} (P : A → Set ℓ₂) {x y : A} → x ≡ y → P x → P y
subst P refl px = px

```

Next comes the means to say that something *cannot* be. A guard against collapse is worthless unless impossibility can be expressed in a way the checker will enforce. The empty type has no constructors and therefore no inhabitants; from a merely supposed inhabitant of it, anything whatever follows. Negation is therefore simply the map into impossibility, and inequality — the formal shape of “kept apart” — is the negation of equality.

```

data ⊥ : Set where

⊥-elim : {ℓ : Level} {A : Set ℓ} → ⊥ → A
⊥-elim ()

¬_ : {ℓ : Level} → Set ℓ → Set ℓ
¬ A = A → ⊥

infix 2 _≠_
_≠_ : {ℓ : Level} {A : Set ℓ} → A → A → Set ℓ
x ≠ y = ¬ (x ≡ y)

```

Then comes the means to say “one of two, and which one”: the tagged alternative. An untagged choice would merge its two cases and thereby destroy the very ability to tell them apart. The disjoint union keeps the tag, so that a later case analysis can ask from which side a witness came.

```

infixr 2 _⊔_

data _⊔_ {ℓ : Level} (A B : Set ℓ) : Set ℓ where
  inj₁ : A → A ⊔ B
  inj₂ : B → A ⊔ B

```

Finally comes the means to carry a thing together with a statement about it: the dependent pair, with the ordinary product as its constant special case. It is what lets an existential claim become an object one holds rather than a sentence one merely utters — a witness and a proof about it, travelling together.

```

infixr 3 _×_

record _×_ (A B : Set) : Set where
  constructor _×_
  field
    fst : A
    snd : B

open _×_ public

record  $\Sigma$  (A : Set) (B : A → Set) : Set where
  constructor _Σ_
  field
    fst : A
    snd : B fst

open  $\Sigma$  public

```

With this much in place — equality, impossibility, alternative, and witness — the certificate of a stable distinction can be written as a single object. What does this smallest object look like, the one that carries separation and cover together?

Chapter 9

Distinction as First Formal Trace

A record, in this language, bundles several pieces of data into one object, so that to hold the object is to hold all of them at once. The certificate we described in words — keep the two apart, and leave nothing out — becomes a record with exactly five fields.

```
record Distinction : Set1 where
  field
    S      : Set
    ℓ      : S
    r      : S
    ℓ≠r    : ℓ ≠ r
    cover  : (x : S) → (x ≡ ℓ) ⊔ (x ≡ r)

open Distinction public
```

Read the fields back into the prose that produced them. S is the carrier, the field in which the positions live. The next two, ℓ and r , are the two positions themselves — “left” and “right,” names with no content beyond being two. The fourth, $\ell \neq r$, is separation: a witness that the two positions are not equal, the explicit guard against collapse. The fifth, cover , is exhaustion: a function that, handed any point of the carrier whatsoever, returns a proof that it is one of the two named positions — the explicit guard against surplus. There is no sixth field. The object carries the two dangers’ two answers, and nothing more.

The first thing the record yields is almost too plain to call a result, yet everything downstream leans on it. The cover field, read as a statement, says that the carrier has exactly two classes: every point reduces to ℓ or to r .

law1-1-cover : $(d : \text{Distinction}) \rightarrow (x : \text{S } d) \rightarrow (x \equiv \ell \ d) \uplus (x \equiv r \ d)$
 law1-1-cover = cover

This two-class coverage becomes powerful the moment it is turned from a fact into a tool. If every point is one of the two, then to know something about *all* points it suffices to know it about the two. That is the carrier’s eliminator: given a property that holds at ℓ and at r , the cover carries it to every point, because there is no third case left to inspect.

Distinction-elim :
 $(d : \text{Distinction}) \rightarrow$
 $\{P : \text{S } d \rightarrow \text{Set}\} \rightarrow$
 $P \ (\ell \ d) \rightarrow$
 $P \ (r \ d) \rightarrow$
 $(x : \text{S } d) \rightarrow$
 $P \ x$
 Distinction-elim $d \ \{P\} \ p\ell \ pr \ x \text{ with cover } d \ x$
 $\dots \mid \text{inj}_1 \ x \equiv \ell = \text{subst } P \ (\text{sym } x \equiv \ell) \ p\ell$
 $\dots \mid \text{inj}_2 \ x \equiv r = \text{subst } P \ (\text{sym } x \equiv r) \ pr$

The “no hidden third” we argued for in prose is now an operation the machine performs. There is literally nowhere else for a point to be: the cover hands back one of two tags, and the eliminator has a branch for each. Surplus is not refuted by argument; it is eliminated by construction.

The fourth field does the opposite work. Where the cover forbids a hidden third, separation forbids a secret collapse of the two into one. The field $\ell \neq r$ is precisely what makes identifying the two positions impossible: any attempt to treat them as equal would produce an inhabitant of the empty type, which is to say a contradiction the checker rejects. Collapse, like surplus, is not talked out of existence. It is made impossible to write down.

With collapse forbidden and surplus eliminated, the carrier is no longer two labels of uncertain honesty. It is a genuine, stable two. The next chapter asks what such a two can do — what maps it admits of itself — and finds, at the end of a short and entirely mechanical analysis, the first closed structure of the book.

Chapter 10

What the First Trace Already Yields

Once the carrier is pinned to two classes, the maps from it to itself lose almost all their freedom. Such a map — an endomorphism — is fixed entirely by what it does to ℓ and to r , and each of those can only go to ℓ or to r . To compare two maps the natural relation is pointwise equality: agreement at every argument. On this carrier the relation is especially tractable, since the cover reduces every argument to one of the two boundary cases.

```
infix 4  $\overset{\circ}{=}$ 
 $\overset{\circ}{=}$  : { $\ell_1 \ell_2$  : Level} { $A$  : Set  $\ell_1$ } { $B$  : Set  $\ell_2$ }  $\rightarrow (A \rightarrow B) \rightarrow (A \rightarrow B) \rightarrow$  Set ( $\ell_1 \sqcup \ell_2$ )
 $\overset{\circ}{=}$  { $A = A$ }  $f g = (x : A) \rightarrow f x \equiv g x$ 

id : { $A$  : Set}  $\rightarrow A \rightarrow A$ 
id  $x = x$ 
```

Among the maps one is quietly special: the map that exchanges the two positions, sending ℓ to r and r to ℓ . Call it the dual. Applied twice, it returns every point to where it started — it is its own inverse, the single nontrivial symmetry a bare two-point carrier possesses.

```
Distinction-dual : ( $d$  : Distinction)  $\rightarrow$  S  $d \rightarrow$  S  $d$ 
Distinction-dual  $d x$  with cover  $d x$ 
... | inj1 _ = r  $d$ 
... | inj2 _ =  $\ell$   $d$ 

Distinction-dual-involutive :
( $d$  : Distinction)  $\rightarrow$ 
( $x$  : S  $d$ )  $\rightarrow$ 
Distinction-dual  $d$  (Distinction-dual  $d x$ )  $\equiv x$ 
Distinction-dual-involutive  $d =$ 
```

```

Distinction-elim d proof-ℓ proof-r
where
proof-ℓ : Distinction-dual d (Distinction-dual d (ℓ d)) ≡ ℓ d
proof-ℓ with cover d (ℓ d)
... | inj₂ ℓ ≡ r = ⊥-elim ((ℓ ≠ r d) ℓ ≡ r)
... | inj₁ _ with cover d (r d)
... | inj₁ r ≡ ℓ = ⊥-elim ((ℓ ≠ r d) (sym r ≡ ℓ))
... | inj₂ _ = refl

proof-r : Distinction-dual d (Distinction-dual d (r d)) ≡ r d
proof-r with cover d (r d)
... | inj₁ r ≡ ℓ = ⊥-elim ((ℓ ≠ r d) (sym r ≡ ℓ))
... | inj₂ _ with cover d (ℓ d)
... | inj₂ ℓ ≡ r = ⊥-elim ((ℓ ≠ r d) ℓ ≡ r)
... | inj₁ _ = refl

```

Now the central question: how many maps are there, counted up to pointwise equality? The boundary table allows exactly four combinations, one for each pair of choices at ℓ and r — the constant-left map, the constant-right map, the identity, and the dual. We name the four once and for all.

```

data EndoCase : Set where
case-constL : EndoCase
case-constR : EndoCase
case-id     : EndoCase
case-dual   : EndoCase

```

The four names mean nothing until they are tied to actual maps and the tie is proved. Inside a module parametrized by a fixed distinction, two maps cross between labels and functions: `interpret` turns each label into its endomorphism, and `classify` inspects an endomorphism's two boundary values and returns the matching label.

```

module K₄ (d : Distinction) where
Endo : Set
Endo = S d → S d

≡-refl : {f : Endo} → f ≡ f
≡-refl x = refl

≡-sym : {f g : Endo} → f ≡ g → g ≡ f
≡-sym p x = sym (p x)

≡-trans : {f g h : Endo} → f ≡ g → g ≡ h → f ≡ h
≡-trans p q x = trans (p x) (q x)

constL : Endo
constL _ = ℓ d

constR : Endo

```

```

constR _ = r d

dual : Endo
dual = Distinction-dual d

dual-ℓ : dual (ℓ d) ≡ r d
dual-ℓ with cover d (ℓ d)
... | inj1 _ = refl
... | inj2 ℓ≡r = ⊥-elim ((ℓ≠r d) ℓ≡r)

dual-r : dual (r d) ≡ ℓ d
dual-r with cover d (r d)
... | inj1 r≡ℓ = ⊥-elim ((ℓ≠r d) (sym r≡ℓ))
... | inj2 _ = refl

interpret : EndoCase → Endo
interpret case-constL = constL
interpret case-constR = constR
interpret case-id     = id
interpret case-dual   = dual

classify : Endo → EndoCase
classify f with cover d (f (ℓ d)) | cover d (f (r d))
... | inj1 _ | inj1 _ = case-constL
... | inj2 _ | inj2 _ = case-constR
... | inj1 _ | inj2 _ = case-id
... | inj2 _ | inj1 _ = case-dual

```

Reading a function's label and then interpreting it returns the same function: at the two boundary points by direct inspection, and therefore everywhere by the eliminator. The two boundary lemmas this rests on are routine case analyses; they are part of the checked file and are not reprinted here.

```

classify-sound : (f : Endo) → interpret (classify f) ≐ f
classify-sound f x = Distinction-elim d (sound-at-ℓ f) (sound-at-r f) x

```

The label is not merely available; it is forced. No two distinct labels can interpret to the same function — sixteen comparisons, one for each ordered pair of labels, every one discharged by the single separation witness $\ell \neq r$. That case analysis is routine and lives in the checked file without crowding the page. From it, classification is unique, and reading a label out and back in returns the very same label.

```

classify-unique : (f : Endo) → (c : EndoCase) → interpret c ≐ f → c ≡ classify f
classify-unique f c ≐ f =
  interpret-injective c (classify f) (≐-trans c ≐ f (≐-sym (classify-sound f)))

classify-interpret : (c : EndoCase) → classify (interpret c) ≡ c
classify-interpret c = sym (classify-unique (interpret c) c ≐ refl)

```

The last line is the small return the whole journey was after. Pull the structure of maps apart into labels with `classify`, push the labels back into maps with `interpret`, and you arrive exactly where you began. The four labels, finally, are genuinely four — pairwise distinct, no two secretly equal — which the checker certifies by the impossibility of equating any two constructors.

```
EndoCase-distinct :
(case-constL ≡ case-constR → ⊥) ×
(case-constL ≡ case-id   → ⊥) ×
(case-constL ≡ case-dual → ⊥) ×
(case-constR ≡ case-id   → ⊥) ×
(case-constR ≡ case-dual → ⊥) ×
(case-id ≡ case-dual     → ⊥)
EndoCase-distinct =
(λ (),
 ((λ ()),
  ((λ ()),
   ((λ ()),
    ((λ ()),
     (λ ()))))))
```

Gathered into two top-level statements, the result reads cleanly: every endomorphism of a distinction is one of the four cases (soundly), and is so in exactly one way (uniquely).

```
k4-classification-sound :
(d : Distinction) →
(f : S d → S d) →
Σ EndoCase (λ c → K4.interpret d c ≐ f)
k4-classification-sound d f = K4.classify d f , K4.classify-sound d f

k4-classification-unique :
(d : Distinction) →
(f : S d → S d) →
(c1 c2 : EndoCase) →
K4.interpret d c1 ≐ f →
K4.interpret d c2 ≐ f →
c1 ≡ c2
k4-classification-unique d f c1 c2 p1 p2 =
K4.interpret-injective d c1 c2 (K4.≐-trans d p1 (K4.≐-sym d p2))
```

A word of care, because it is easy to over-read the number four. These four maps are closed under composition, but they do not form a group. The identity and the dual, taken by themselves, do: each is its own inverse, a two-element group. The two constants, however, are not invertible — a constant map composed with anything is again that same constant — so the four together form a monoid, not a group. What the checker establishes is the count and the classification: exactly four endomorphisms, sound and unique, up to pointwise equality. Any

richer algebraic reading of those four — as the vertices of a particular finite group, say — is a further step, taken in the open, and not part of what has just been proved.

And here, precisely, is the footprint we set out to find — not the origin, and not the destination, but the first place where a single step of the long road — the step from “two positions, kept apart, with nothing left over” to “a closed, four-element structure that returns to itself” — becomes exact enough for a machine to check without any appeal to intuition. The road did not begin here. It arrives here. What remains is to turn around and ask what such a footprint is worth.

Chapter 11

What Has Actually Been Shown?

A checked threshold is not a destination; it is a vantage. Having reached it, we can turn around and say, with more precision than was available before, exactly what has been secured and exactly what has not.

What the machine now holds is small and definite: a carrier with two positions, a witness that they are apart, an exhaustion principle leaving no third case, and exactly four self-maps up to pointwise equality, with a return from maps to labels and back again. Under `--safe` and `--without-K`, each of these is theorem in the strict sense: if any one failed, the file would not compile.

It is just as important not to hear more in these theorems than they say. They do not prove that the world was deduced from nothing, nor do they prove the earlier claim that description presupposes distinction. That claim concerns what any formalization already stands on, so it cannot be returned as one of formalization's own results. What the threshold shows is narrower and more exact: once two positions are granted in full view and kept apart, that minimal grant already carries a closed shape. Agda is one disciplined carrier of that result, not a proof that no other carrier could exist; whether the same content can be checked elsewhere remains open. The metaphysical claim is the reason one cares about the threshold; it is not among the things the threshold proves.

The claim earned here can therefore be stated in one sentence. The checked closure of binary distinction is the first stable skeleton of formulability: the least a system must carry to host a single stable distinction, exhibited as a normal form, a closed four-element endomorphism ledger, and a return to its own datum. It

does not derive the world. It marks the point at which this minimal performance becomes checkable. But how far does such a foothold carry back toward physics?

Chapter 12

Toward Form

The book set out from physics, at the edge where the cosmic picture stops being a picture. It is owed a return to that horizon. But the return must be made under a discipline, because the distance between a checked theorem and a measured quantity is exactly the place where ambitious projects lose their honesty.

Three courts decide three different questions: the compiler for the formal kernel, ordinary mathematics for the derivations around it, and experiment for any reading of the kernel as physics. One ledger, three judges.

Read in that ledger, the threshold reached in this book is the first rung, and its status is unambiguous: it is theorem. That the self-maps of a stable distinction number exactly four, up to pointwise equality, is arithmetic the kernel forces. Read the four-element closure as the corners of a small complete figure and the count of its pairings, $\binom{4}{2} = 6$, is again arithmetic — computed, and checked. Nothing physical has been claimed at this rung. These are integers, and they are theorems about integers.

The larger books continue that arithmetic. A polynomial in the closure's invariants, for instance, yields 137. As arithmetic, that too is theorem. Only when one reads such an integer as a constant of nature does theorem give way to structural hypothesis, and only measurement can decide the comparison. If a measurement fails, the identification fails; the arithmetic remains what it was.

So the question of origin can be returned to its horizon, changed but not answered. Physics still cannot describe its own starting point; the edge at the beginning of the picture is still an edge. What has changed is narrow. The first step behind the starting point — the holding-apart of one position from another — is no longer only a matter of intuition. It has a checked normal form, a closed endo-

morphism ledger, and a return to its own datum. We have not crossed the edge. But directly at it, a small exact foothold has been gained.

Conclusion — The First Footprint

Return, one last time, to the film running backward: galaxies drawing together, the universe growing hotter and denser, the whole picture contracting toward a single brilliant instant — and then, before the first frame, the edge, where the account stops being an account.

We began by doubting that the edge was only a shortage of data, and we followed the doubt past cosmology, past Leibniz's question of why there is something rather than nothing, to a suspicion about description itself: that before anything can be measured, recorded, or even mistaken, something must already stand apart from something else. That suspicion is not a theorem, and this book has been careful never to dress it as one. What the book could nevertheless do is tightly bounded. It could follow that pressure to the first place where a single step becomes exact and write down there a datum that a machine can check.

It matters that the footprint remain a footprint and not become a hero. The distinction we formalized is not the origin, not the ground of being, not the answer to Leibniz. It is the earliest trace of a road that was already long underfoot when we reached it — and that is precisely why the journey begins where it does. Not because distinction is the answer, but because it is the first place the road becomes checkable. A book that had opened by announcing distinction as its foundation would have inverted exactly the order that matters to it. The order runs the other way: from the question of origin, through what every determination already presupposes, through the informational pressure that asks what distinction must be able to perform, and only then down to the small checked record in which the first stretch of the road can finally bear weight. Here, in strict form, there returns what was already latent in Leibniz through the contrast between something and nothing.

That is the whole of the offer, and it is meant to be taken at exactly its weight — no less, and no more. From “why is there something rather than nothing” — a question that already sets *something* against *nothing* — to a record carrying a separation and a cover, one single thread runs unbroken, and at its far end a compiler accepts the first exact step. The edge of the picture is still there. But the question of origin now has, for the first time, a small and exact foothold on a road that thinkers have been walking for a very long time.

The road did not begin at the distinction.

It arrived there — and there, for the first time, it could be checked.

Outlook to Void and Form

This book stops where the first exact foothold appears. The larger books ask what happens if that foothold is not merely named but allowed to run. In the hard Leibnizian mood — in the old phrase, if God does not play dice — the demand is not for one elegant option among many. It is for the cutting away of option after option until only what the formal data force remains.

That mood was physical from the beginning, and it is worth saying why. The question this companion opened with was never purely abstract. It stood at the edge of the cosmic picture, at the first instant the backward film cannot reach; and to ask why there is something rather than nothing is already to stand at that same edge. The search for a first distinction was driven, from the start, by a suspicion that the place where physics loses its own beginning and the place where description loses its own ground might be one place. That suspicion was the motor. It is why the road runs toward *Void* and *Form* rather than stopping at a tidy piece of logic: the logic was only ever the disciplined way of testing a thought about the world.

One picture made the suspicion feel concrete; it is offered as intuition, not as proof. Suppose a single blank sheet lies on a table. That one sheet still marks no difference. Only when a second sheet is added does this rather than that become available at all. The two may not collapse, or there would again be only one. And if no third sheet is to remain unaccounted for, the field must also be closed completely.

In this little image, the two later demands already lie together: separation against collapse and cover against surplus. The distinction record later writes precisely these two data down formally. Spencer Brown's simplest step returns here: a distinction is drawn. And where this-rather-than-that becomes a held difference, the same thought returns that Wheeler later reads as bit. Its closure yields K_4 not as metaphor, but as a dependency result.

But the reader should not be asked to leap in one motion from this small checked record to the constants of nature. Too much changes across that distance. The next step is therefore not: now physics is proved. The next step is: now the pressure that brought us to the record is allowed to keep working inside a larger formal development, and we watch what survives.

That enlargement happens in stages. First the formal kernel is extended without changing its discipline: the same distinction is pursued through normal form, classification, closure, dependency, arithmetic, and later the construction of a continuous arena where missing limits can live. Only after that enlarged mathematical field exists does the question of physical reading become legitimate. The order matters. If one rushes straight from the record to the world, the project becomes a manifesto. If one lets the formal pressure keep doing its work, the larger claim at least arrives with a trace of discipline behind it.

That is what *Void* investigates. It begins from collapse pressure: if two alleged positions cannot be distinguished by anything expressible in the system, the system has no internal ground for holding them apart. To stabilize two positions against that pressure, the formal datum must carry anti-collapse data and anti-surplus data: separation, so the two positions do not identify, and cover, so no hidden third point survives in the carrier. From there the development tightens mechanically. Every separated and covered binary carrier has normal form Two; its endomorphism space has exactly four cases; under the stated representation conditions the graph closure is K_4 ; and the closure returns to the originating Distinction datum. No postulates are added, no stronger equality principle is imported, and no hidden case is left outside the check.

That is why *Void* matters. It does not treat distinction as a static symbol. It lets distinction run as a process inside a strict formal environment and watches what freedom is eliminated at each step. The result is not a metaphor but a dependency chain: anti-collapse, anti-surplus, normal form, four-case classification, closure, and return. Even the arithmetic needed later is built inside the same file: the natural numbers, integers, rationals, and Cauchy reals are constructed in Agda rather than borrowed from outside. So when the later ledger is read, it is not being painted onto the kernel from above. It is being computed from within the same checked development.

That Cauchy step matters far beyond bookkeeping. In *Void*, the metric on $\mathbb{Q}$ is incomplete: Cauchy data cannot remain inside the rationals without loss, so the reals are forced as the Cauchy closure of $\mathbb{Q}$. Inside this framework, that is the

precise passage by which a discrete ledger acquires a legitimate smooth arena. Continuity is not laid around the argument as atmosphere. It appears only where the missing limits make it unavoidable.

There is a concrete image for the same step. The corners of the closure carry only whole numbers; the first value that is not a whole number appears at the centroid of the tetrahedron, the interior point whose barycentric coordinates $(1/4, 1/4, 1/4, 1/4)$ average the four corners and so force the denominator 4. That interior point is where the rationals enter, and the refinement limits taken over them are where the reals enter. Without an inside to the figure the ledger could speak only in integers; the centroid is the smallest step by which it can begin to carry decimals at all — which is exactly what any comparison with measured constants will require.

Only after all this does the next surprise come into view. The running of distinction does not merely produce one more theorem of the same size. It leaves behind a finite invariant record that can be read, before any interpretation begins, as a sealed ledger of integers, multiplicities, and structural constraints. At that point the question is still mathematical: what does the closed record contain? Only later does *Form* ask whether the world might be reading from the same ledger.

Once the closure is in place, a small invariant record can be read off: $(V, E, d, \chi, \kappa) = (4, 6, 3, 2, 8)$. From this record come the first entries of the ledger. The simplex evaluation $V^d \cdot \chi + d^2$ is, within *Void*'s expression grammar, the uniquely admissible form, and on K_4 it yields 137. The Clifford branch yields the Fermat prime $F_2 = 17$. From the same invariant stock come the tree-level mass formulae $\chi^2 \cdot d^3 \cdot F_2 = 1836$, $d^2 \cdot (E + F_2) = 207$, and $F_2 = 17$, together with the structural readings $d = 3$ spatial dimensions, $V - d = 1$ time dimension, and three degenerate Laplacian modes. The graph is no longer merely a closure object. It has become a ledger.

That ledger is what *Form* learns to read. Here the boundary that has governed this whole companion becomes decisive. *Void* never says that 137 is the inverse fine-structure constant, that 1836 is the proton-to-electron mass ratio, or that the threefold degeneracy is three generations of matter. *Void* proves integers and structural facts. *Form* performs the act of identification. That is the largest change in scale anywhere in this companion, and it is why the transition has to be slow. One is no longer asking what the formal object forces internally, but whether nature's open ledger might coincide, in part, with the closed ledger

the formal chain has produced. There the remarkable part begins: 137 is read against $\alpha^{-1} = 137.035\,999\,084(21)$, exact at the integer level with a fractional residual of about 0.036%; 1836 is read against the measured proton-to-electron ratio 1836.15267343(11); 207 and 17 enter as the tree-level integer slots for the muon/electron and tau/muon ratios; the same K_4 arithmetic is read structurally as $3 + 1$ spacetime, three generations, and the force-carrier count $V \cdot d = 12$. At that point the scale of the project changes.

Nor does *Form* stop at the integer shell. The later decimal places are not recovered by tuning. They are approached through a correction chain whose arithmetic is itself constrained by the same ledger: a universal loop numerator $11 = E + d + \chi$, a bulk correction $+11/72$, an extended correction $-11/576$, a muon bridge $-16/69$, a tau extension $-28/153$, an electromagnetic correction $+11/306$, and the baryonic closure $8/51$. The book presents these sectors through dedicated classification records and then assembles them into a single master correction record carrying the relevant numerators, denominators, and uniqueness witnesses. Even the orientation of the first two scales is fixed: bulk and extended role cannot be swapped without breaking the realized equations. The one irreducibly interpretive step is not the fractions themselves but the assignment of physical names to already realized fractions. That is how the larger argument tries to move from exact integers toward measured decimal precision without re-opening free choice.

But it changes under a discipline, not under intoxication. In *Form*, the separation between formal result, forced arithmetic, physical identification, and experimental test is itself written as a machine-verified record. The book keeps saying which verdict belongs to which court because otherwise the whole program would collapse into overclaim. The integers are not negotiable relative to the checked kernel. The identifications are. That is why the larger work can be bold without becoming vague.

It is also why the larger work can be killed cleanly by the world. The K_4 arithmetic carries no free parameters, no couplings to tune, and no hidden sectors to absorb failure. If the inverse fine-structure constant had settled at 136 or 138 rather than 137; if the proton-to-electron ratio were nowhere near the 1836 neighborhood; if a fourth spatial dimension or a fourth fermion generation turned up; if the baryonic correction chain did not land where *Form* says it must land — then the physical reading would fail. The mathematics would remain; the identification would collapse. That is not a weakness. It is the minimum honesty

required of a claim this large.

And this is the point at which the emotional scale changes as well. Up to here, one may reasonably feel: the first question is real, the route is clean, the threshold is not trivial, and something may be there. *Void* and *Form* are where that feeling is put under maximal pressure. They ask whether the same road that led from Leibniz's question to the first checked distinction can continue, mechanically and without fresh arbitrariness, into the integer ledger physics still takes as given. If that program holds, even partially, then the old metaphysical pressure has begun to cast a shadow across the open constants of the world. This companion does not prove that scale. It shows where it begins.